

SIMATS ENGINEERING
CO2_ASSESSMENT TOOL 1: Analytical Problem Solving

COURSE NAME: Cloud Computing and
Big Data Analytics

COURSE CODE: CSA1522

Name	DHILIBAN M
Register Number	192421166

COURSE OUTCOME COVERED

CO2: *Analyze virtualization architectures to identify performance and security issues in cloud computing environments. (BL3)*

Dave's Taxonomy: Manipulation (Level 2)

Total Marks: 50

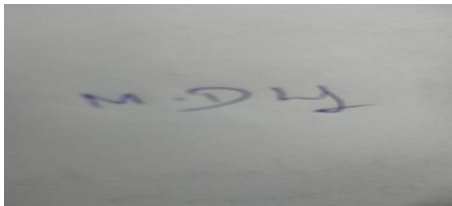
Question Count: 5

Code of Conduct

I DHILIBAN M/192421166 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

A photograph of a piece of paper with the name 'DHILIBAN M' written in blue ink, appearing to be a signature or stamp.

Assessment Tasks

Analytical Problem Solving

1. Compare Type-1 and Type-2 hypervisors for a cloud datacenter hosting mission-critical applications. Analyze performance, security, and manageability, then justify the better choice.
2. A cloud provider must isolate workloads belonging to finance, healthcare, and student portals on the same hardware. Analyze how virtualization architecture supports isolation and identify two major attack surfaces.
3. Study the following VM host utilization snapshot and recommend corrective actions: CPU 92%, memory 88%, storage I/O latency 35 ms, average VM boot time 170 s. Explain the likely bottlenecks.

4. Analyze how Software Defined Datacenter architecture improves agility compared with a traditional datacenter. Support your answer with compute, storage, and network virtualization considerations.
5. A multi-cloud federation project fails due to identity mismatches between providers. Analyze the issue and propose a secure identity and privacy design using federation concepts.

Analytical Problem Solving Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 – Proficient	Level 2 - Developing	Level 1 - Beginning
Problem Interpretation	20%	Interprets all technical requirements accurately	Interprets most requirements correctly	Partial interpretation	Misinterprets the problem
Analytical Reasoning	25%	Strong reasoning with clear cause-effect analysis	Good reasoning with minor gaps	Basic analysis only	Weak or no analysis
Method Selection	20%	Chooses highly appropriate virtualization and security concepts	Chooses mostly suitable concepts	Partially suitable concepts	Inappropriate approach
Solution Quality	20%	Provides feasible and technically sound solution	Reasonable solution with minor issues	Basic solution with limitations	Unclear or invalid solution
Presentation of Analysis	15%	Well-structured and technically precise	Generally clear	Somewhat unclear	Difficult to follow

Total Marks Awarded:

1. Compare Type-1 and Type-2 hypervisors for a cloud datacenter hosting mission-critical applications. Analyze performance, security, and manageability, then justify the better choice.

Performance:

Type-1 (bare-metal) runs directly on hardware, offering near-native performance and lower latency.

Type-2 (hosted) runs on top of an OS, introducing overhead and reduced efficiency.

Security:

Type-1 has a smaller attack surface since it eliminates the host OS layer.

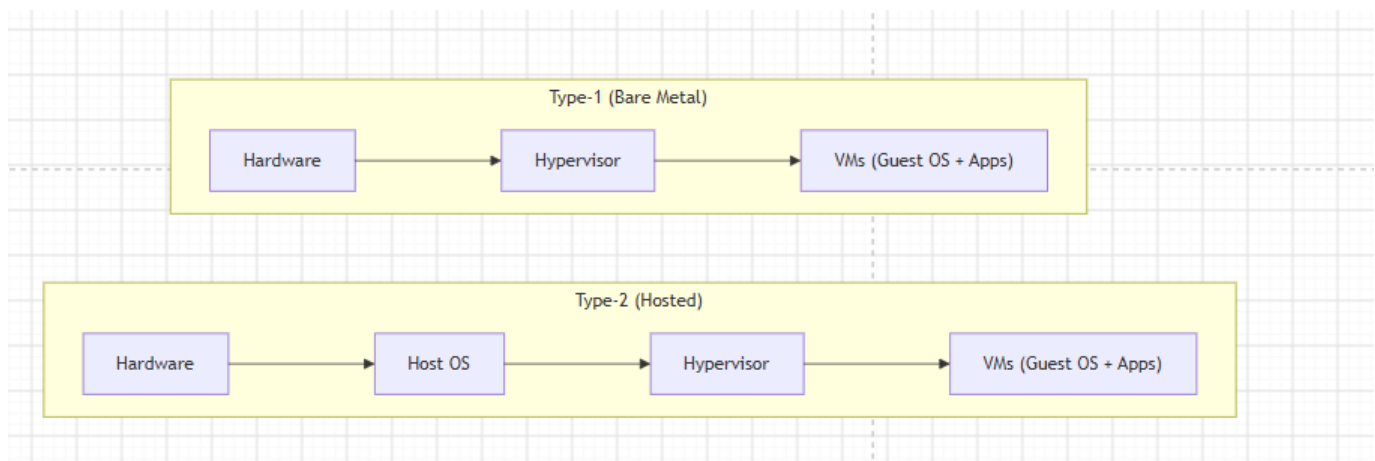
Type-2 inherits vulnerabilities from the host OS, making it less secure for mission-critical workloads.

Manageability:

Type-1 integrates with enterprise tools (VMware ESXi, Hyper-V) for centralized management.

Type-2 is simpler for desktops but lacks enterprise-grade orchestration.

Justification: For a cloud datacenter hosting mission-critical applications, Type-1 hypervisors are superior due to performance, hardened security, and enterprise manageability.



2.A cloud provider must isolate workloads belonging to finance, healthcare, and student portals on the same hardware. Analyze how virtualization architecture supports isolation and identify two major attack surfaces.

Isolation Mechanism:

Hypervisors enforce strong logical separation of VMs, ensuring finance, healthcare, and student workloads cannot access each other's memory or CPU states.

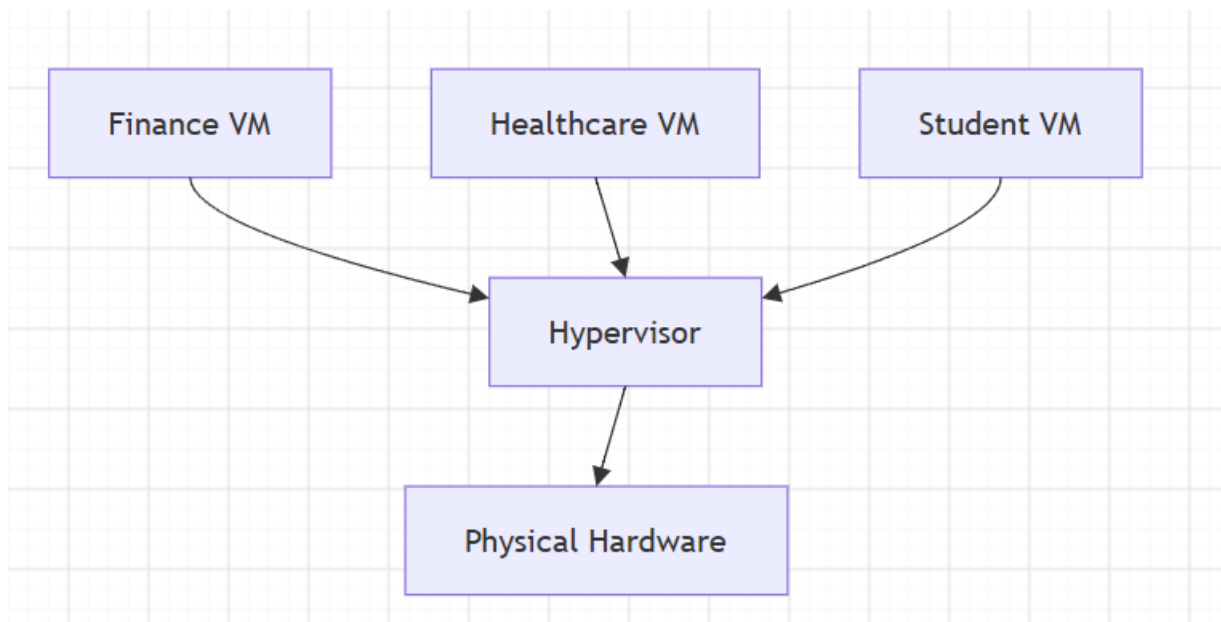
Virtual networking (VLANs, VXLANs) and storage segmentation further isolate workloads.

Major Attack Surfaces:

Hypervisor exploits (e.g., VM escape attacks).

Shared resources (e.g., side-channel attacks via CPU cache or I/O).

Conclusion: Virtualization provides workload isolation, but securing the hypervisor and monitoring shared resource vulnerabilities is critical.



3.Study the following VM host utilization snapshot and recommend corrective actions: CPU 92%, memory 88%, storage I/O latency 35 ms, average VM boot time 170 s. Explain the likely bottlenecks.

Metrics:

CPU 92% → Overcommitment or runaway processes.

Memory 88% → Risk of swapping, ballooning overhead.

Storage I/O latency 35 ms → Indicates disk bottleneck (should be <5 ms).

VM boot time 170 s → Slow due to I/O contention and high CPU load.

Corrective Actions:

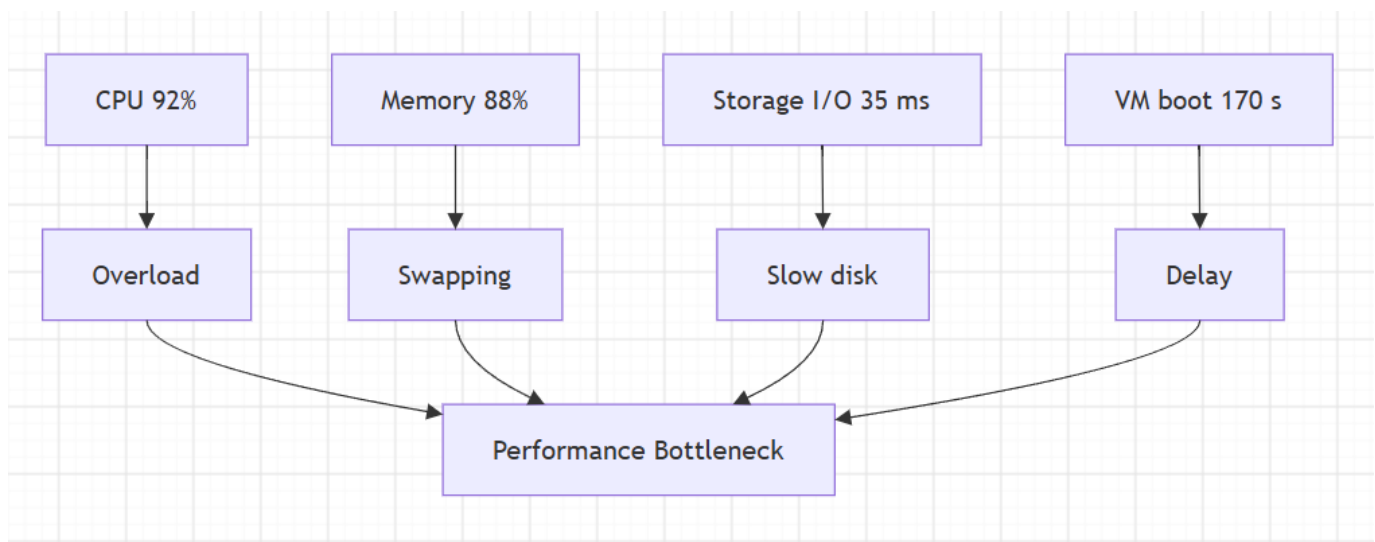
Migrate VMs to other hosts (load balancing).

Add CPU/memory capacity or enable auto-scaling.

Upgrade to SSD/NVMe storage or optimize I/O scheduling.

Implement resource reservations for critical VMs.

Likely Bottlenecks: Storage I/O and CPU saturation are the primary causes of degraded performance.



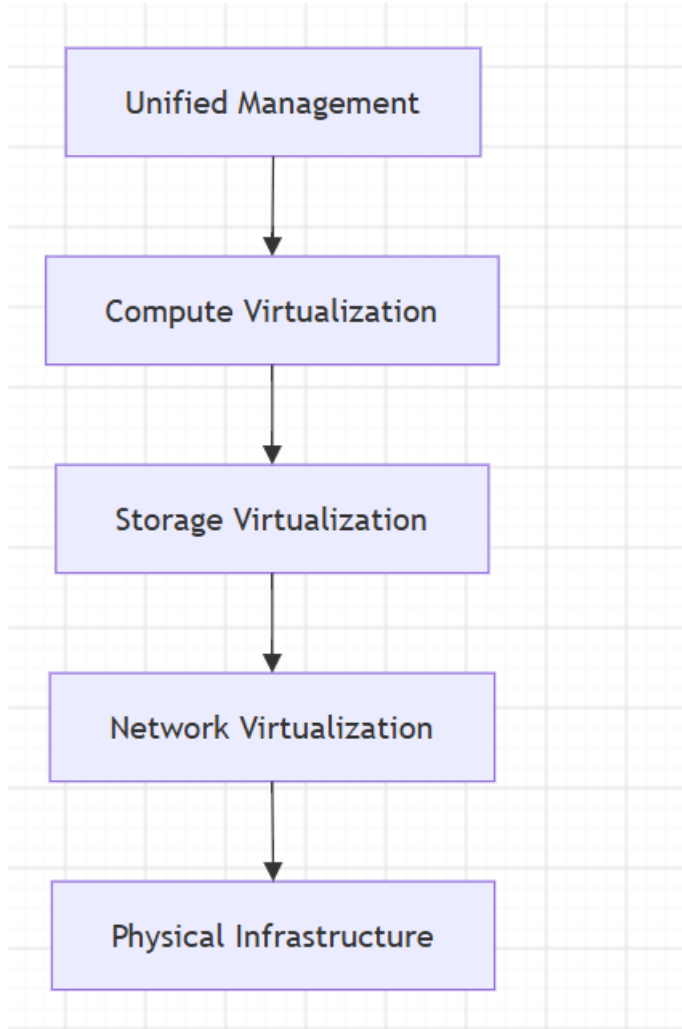
4. Analyze how Software Defined Datacenter architecture improves agility compared with a traditional datacenter. Support your answer with compute, storage, and network virtualization considerations.

Compute Virtualization: Rapid provisioning of VMs/containers compared to manual server deployment.

Storage Virtualization: Abstracts physical disks into pools, enabling dynamic allocation and thin provisioning.

Network Virtualization: Virtual switches, overlays (VXLAN) allow instant reconfiguration without physical rewiring.

Agility Gain: SDDC enables automated orchestration, elastic scaling, and policy-driven management, unlike traditional datacenters that rely on manual hardware provisioning.



5.A multi-cloud federation project fails due to identity mismatches between providers. Analyze the issue and propose a secure identity and privacy design using federation concepts.

Issue: Different providers use distinct identity formats (Azure AD, AWS IAM, Google Identity), leading to mismatches in authentication and authorization.

Solution Design:

Use Federated Identity Management (FIM) with standards like SAML, OAuth 2.0, and OpenID Connect.

Implement a central Identity Provider (IdP) that issues tokens trusted across clouds.

Apply privacy-preserving techniques (minimal attribute disclosure, encrypted assertions).

Outcome: A secure federation ensures seamless cross-cloud authentication while protecting user privacy.

